



Image Analysis Collaboratory



ACITE

Core for Imaging Technology & Education

Bit Depth

The **bit depth** defines the range of possible **digital values** that each pixel can have, usually **8, 12 or 16 bit**.

The **bit depth** is expressed in **grey values**.

bit depth of the image = bit depth of the detector

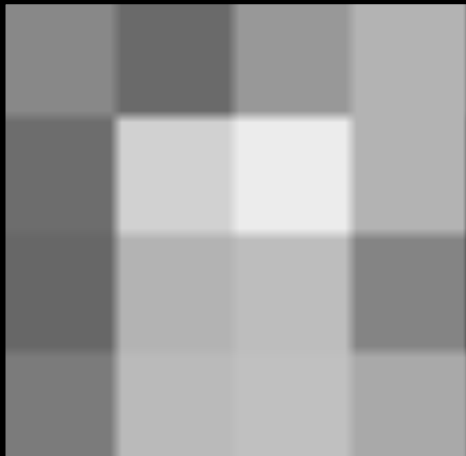
(Unless you change that during acquisition)

x bit = a range of 2^x grey values

8 bit image = **each pixel** can have **2^8 grey values** = 256 grey values = **range 0-255**

12 bit image = **each pixel** can have **2^{12} grey values** = 4096 grey values = **range 0-4095**

16 bit image = **each pixel** can have **2^{16} grey values** = 65536 grey values = **range 0-65535**





136	106	152	179
109	209	236	179
103	179	189	132
123	186	192	169

Digital Value = Pixel Intensity Value = Grey Value

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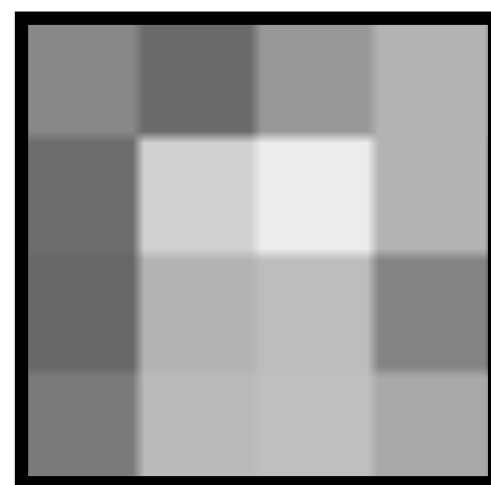
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display your images